

5.7 Metapopulation Model

5.7.1 Levins Model

Metapopulation means a collection of subpopulations separated in space which have some ecological interactions, for example, by migration between them. This concept has created by an ecological researcher Richard ‘Dick’ Levins (1930–2016) in his work of 1969 [10]. He expressed it as “a population of populations”. Levins [10, 11] proposed also a new mathematical modeling for the population dynamics based on the concept. His idea of the modeling has been widely accepted and applied for a variety of theoretical researches in landscape ecology and conservation biology, especially in the ecological field treating the population dynamics for a large spatial scale.

The essence of Levins’ idea proposed in [10, 11] is to binarize the state of local habitat, briefly *patch*, with respect to the trend of the subpopulation size in it. The possible alternative states are ‘occupied’ and ‘empty’ (or ‘vacant’). This idea ignores the detail of subpopulation size in each patch, and focuses on the establishment of subpopulation at the patch in a certain ecological sense. Hence in other words, it may be regarded as a modeling of *coarse-graining* population dynamics. The metapopulation dynamics in this meaning is to describe the temporal change of the frequency of occupied patches in the whole habitat which consists of a collection of local habitats fragmented in space, that is, habitable patches.

The mathematical model proposed by Levins [10, 11] is as follows:

$$\begin{aligned}\frac{dE}{dt} &= eP - cPE; \\ \frac{dP}{dt} &= cPE - eP,\end{aligned}\tag{5.27}$$

where E and P mean the frequencies of ‘empty’ and ‘occupied’ patches respectively. They may be regarded as the probabilities of such states at an arbitrarily chosen patch. The occupied patch is the local habitat established by a subpopulation in an ecological sense. The empty patch is the local habitat regarded as with no such an establishment, which may be vacant before any settlement, or may be emerged after the extinction of subpopulation previously inhabiting there. Such an extinction may be caused for example, by a stochastic disturbance in the environment with a climate change or human interruption. The parameter e is the coefficient of the occurrence of event to make the patch ‘empty’ (e.g., extinction or abandonment). The parameter c is the coefficient of the settlement to make the patch ‘occupied’. The term cP introduces the likeliness of the occurrence of settlement by the immigration from occupied patches. This indicates that the probability of the settlement to an empty patch is proportional to the frequency of occupied patches in the whole habitat, whereas cP does not have any meaning of probability in itself

(refer to the below arguments in the gray box). The model (5.27) is frequently called *Levins model*.

Let us suppose that there are N patches in the whole habitat for the metapopulation. Then $NP(t)$ and $NE(t)$ respectively define the (expected) numbers of occupied and empty patches at time t . Now, the probability that an empty patch is settled by immigrants in a sufficiently short period $[t, t + \Delta t]$ is assumed to be given by $\rho NP(t)\Delta t + o(\Delta t)$, where the positive parameter ρ is the coefficient reflecting the easiness/hardness of settlement in the empty patch. At the same time, the probability that an occupied patch becomes empty in $[t, t + \Delta t]$ is assumed to be given by $e\Delta t + o(\Delta t)$. Then the change in the frequency of empty patches during $[t, t + \Delta t]$, that is, $NE(t + \Delta t) - NE(t)$ satisfies the following relation:

$$NE(t + \Delta t) - NE(t) = -\{\rho NP(t)\Delta t + o(\Delta t)\}NE(t) + \{e\Delta t + o(\Delta t)\}NP(t).$$

Dividing both sides by $N\Delta t$ and taking the limit as $\Delta t \rightarrow 0$, we can get the first equation of (5.27) with $c = \rho N$.

From the modeling and definition, it is always satisfied that $0 \leq E \leq 1$, $0 \leq P \leq 1$, and $E + P = 1$. Hence, the model (5.27) can be expressed by the following one dimensional ordinary differential equation:

$$\frac{dP}{dt} = cP(1 - P) - eP \quad (5.28)$$

$$= \{(c - e) - cP\}P. \quad (5.29)$$

The Eq. (5.28) is sometimes introduced as Levins model instead of (5.27). As seen from the Eq. (5.29), Levins model (5.28) is mathematically equivalent to the logistic equation (5.10). If $c \leq e$, the frequency of occupied patches P monotonically decreases to asymptotically converge to 0, that is, the population goes extinct (Fig. 5.10a). If $c > e$, the frequency P asymptotically converges to equilibrium $1 - e/c$ (Fig. 5.10b). In this case, the eventual positive frequency of occupied patches means the persistence of population in the habitat. Consequently, it is implied that the population persistence requires a sufficient activity of migration between patches in such a fragmented habitat.

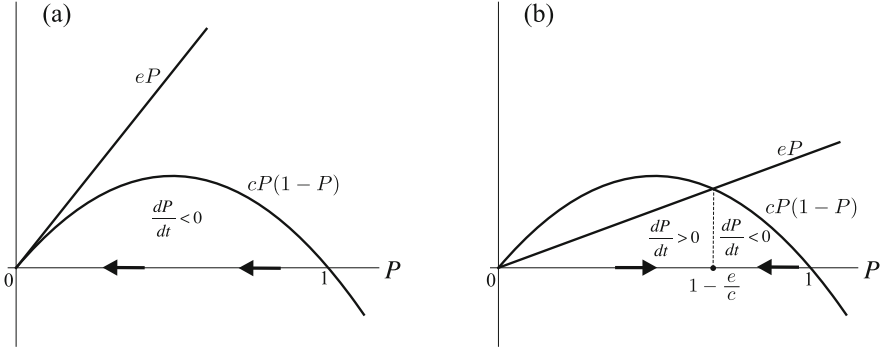


Fig. 5.10 The phase space consideration on the nature of Levins model (5.28). (a) For $c \leq e$, the population goes extinct; (b) For $c > e$, the population asymptotically approaches the persistent equilibrium

5.7.2 3-State Metapopulation Model

A straightforward extension of Levins model (5.27) is by introducing different states of patch in the modeling. In this section, we consider such a model proposed by Ilkka Hanski (1953–2016) in 1985 [5] (see also [6] and [7, p. 61]):

$$\begin{aligned}\frac{dE}{dt} &= e_S S - cLE; \\ \frac{dS}{dt} &= cLE + e_L L - e_S S - rS - mLS; \\ \frac{dL}{dt} &= rS + mLS - e_L L,\end{aligned}\tag{5.30}$$

where E , S , and L are the frequencies of empty patches, occupied patches with a small population size, and occupied patches with a large population size respectively. The state of settled patch is now binarized into two different ones. This modeling provides a metapopulation dynamics with three states of patch. It is satisfied that $E(t) + S(t) + L(t) = 1$ for any time t , which can be reasonably embodied in (5.30) such that $d\{E(t) + S(t) + L(t)\}/dt = 0$ for any time t .

The parameter c is the coefficient of initial settlement by a small subpopulation immigrating in an empty patch. It must be noted that, in (5.30), the migration from patches of the state S is assumed negligible for the settlement. The parameter e_S is the coefficient of the transition to the state ‘empty’ E for the small subpopulation at the state S . The parameter r is the coefficient of the growth of small subpopulation to the large one with the reproduction in the patch, while e_L is the coefficient of the decline of large subpopulation to the small one with some causes to reduce the population size. The parameter m is the coefficient of the increase of subpopulation size at the state S by the immigration from the patches of the state L . The net

immigration rate per small subpopulation is introduced by mL which is proportional to the frequency of large subpopulations. Hanski mentioned in his book [7] that the most important factor in this model is the immigration to small subpopulations from large subpopulations. The effect of such an immigration may be called *rescue effect* for a reason described in the following part.

Since $E(t) + S(t) + L(t) = 1$ for any time t , the system (5.30) is mathematically equivalent to the following two dimensional system:

$$\begin{aligned}\frac{dE}{dt} &= e_S(1 - E - L) - cLE; \\ \frac{dL}{dt} &= (r + mL)(1 - E - L) - e_L L.\end{aligned}\tag{5.31}$$

The extinct equilibrium $(1, 0)$ always exists. By the local stability analysis (refer to Sects. 14.2 and 14.3), we can get the following result on the stability of the extinct equilibrium $(1, 0)$:

$$\left\{ \begin{array}{ll} \text{unstable (saddle)} & \text{if } \frac{c}{e_S} > \frac{e_L}{r}; \\ \text{locally asymptotically stable (stable node)} & \text{if } \frac{c}{e_S} < \frac{e_L}{r}. \end{array} \right.\tag{5.32}$$

First, let us consider the case of $m = 0$ for (5.31), when there is no contribution of the immigration from large subpopulations to the state transition of small subpopulation. For the system (5.31) with $m = 0$, we can easily find that the unique non-trivial equilibrium

$$(E^*, L^*) = \left(E^\dagger, \frac{1 - E^\dagger}{1 + e_L/r}\right)\tag{5.33}$$

exists if and only if $E^\dagger := e_S e_L / (cr) < 1$. Further with the local stability analysis on this equilibrium, we can get the following result about its existence and stability (Exercise 5.4):

$$\left\{ \begin{array}{ll} \text{absent} & \text{if } \frac{c}{e_S} < \frac{e_L}{r}; \\ \text{unique and locally asymptotically stable (stable node)} & \text{if } \frac{c}{e_S} > \frac{e_L}{r}. \end{array} \right.$$

Exercise 5.4 Derive the above result on the local stability of the non-trivial equilibrium (E^*, L^*) given by (5.33) when $m = 0$.

For the system (5.31) with $m > 0$, there is a bistable case with two non-trivial equilibria, E_u and E_s , as illustrated by Fig. 5.11. In such a bistable case, the population goes extinct if the frequency of empty patches is beyond a threshold

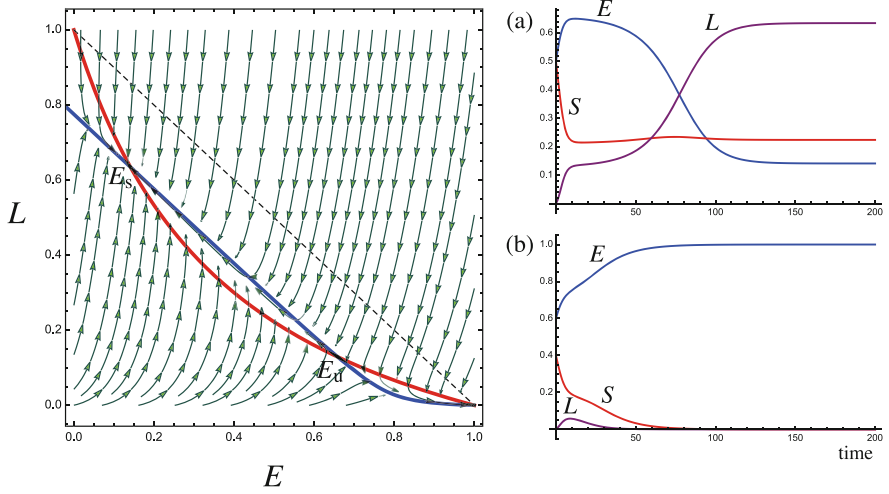


Fig. 5.11 Numerical example of a bistable case for the 3-state metapopulation model (5.30). Nullclines and vector flows in the (E, L) -phase plane (left) and the temporal changes of variables from different initial conditions (right). The dashed line in the (E, L) -phase plane indicates $E + L = 1$. In (a) of right figures, the system asymptotically approaches a persistent state from $(E(0), S(0), L(0)) = (0.5, 0.5, 0.0)$, while in (b), it asymptotically approaches the extinct state from $(E(0), S(0), L(0)) = (0.6, 0.4, 0.0)$. Commonly, $e_s = 0.08$; $c = 0.2$; $e_L = 0.5$; $r = 0.02$; $m = 2.2$

at a certain moment, while it asymptotically approaches a persistent state if the frequency of empty patches is below the threshold.

The existence of such a bistable situation can be illustrated also by the bifurcation diagram as shown in Fig. 5.12. It is indicated that there are threshold values μ_c and ν_c for parameters m/r and c/e_s respectively such that

$$\text{When } \frac{m}{r} \leq \mu_c, \begin{cases} \text{the population goes extinct if } \frac{c}{e_s} < \frac{e_L}{r}; \\ \text{the population approaches the persistent state } E_s \text{ if } \frac{c}{e_s} > \frac{e_L}{r}; \end{cases}$$

$$\text{When } \frac{m}{r} > \mu_c, \begin{cases} \text{the population goes extinct if } \frac{c}{e_s} < \nu_c; \\ \text{a bistable situation appears if } \nu_c < \frac{c}{e_s} < \frac{e_L}{r}; \\ \text{the population approaches the persistent state } E_s \text{ if } \frac{c}{e_s} > \frac{e_L}{r}. \end{cases}$$

The threshold value μ_c is given by $e_L/r + 1$, while ν_c must depend on m/r and e_L/r as seen in Fig. 5.13.

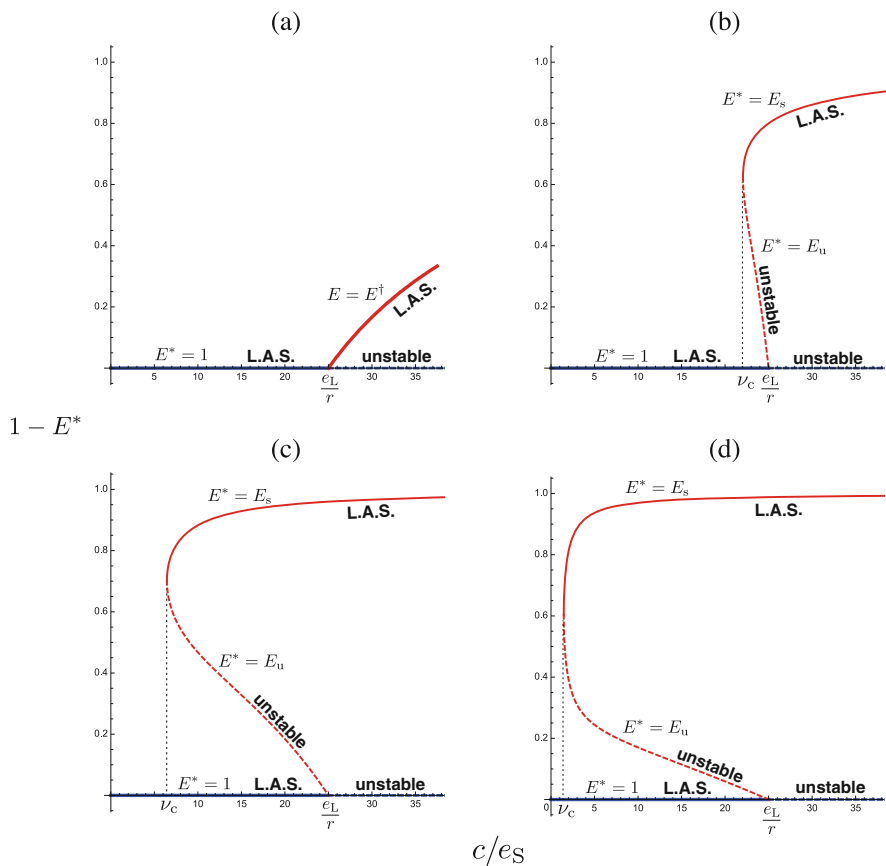
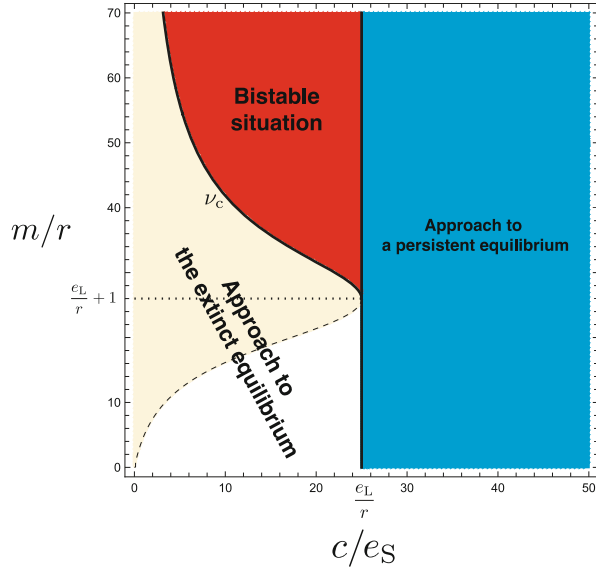


Fig. 5.12 Numerically drawn bifurcation diagrams about the value $1 - E^* = S^* + L^*$ at the equilibrium state for the 3-state metapopulation model (5.30). (a) $m = 0$; (b) $m/r = 30.0$; (c) $m/r = 50.0$; (d) $m/r = 110.0$. Commonly, $e_L/r = 25.0$. In figures, “L.A.S.” means “locally asymptotically stable”

The bifurcation at $c/e_S = e_L/r$ when $m/r \leq \mu_c$ in Fig. 5.12a is classified in what is called *supercritical bifurcation* in the dynamical system theory, at which one equilibrium switches its stability from asymptotically stable to unstable while the other asymptotically stable equilibrium comes to exist at the same time. In contrast, the bifurcation at $c/e_S = e_L/r$ when $m/r > \mu_c$ in Fig. 5.12c–d is called *subcritical bifurcation*. The subcritical bifurcation has a structure called *hysteresis*. On the other hand, the bifurcation at $c/e_S = \nu_c$ is what is called *saddle-node bifurcation*.

Fig. 5.13 Parameter dependence of the asymptotic behavior for the 3-state metapopulation model (5.30). Numerical result for $e_L/r = 25.0$. There are possible three different asymptotic behavior, indicated by thick black solid boundaries



It is clearly indicated by Figs. 5.12a and 5.14a that, when there is no non-trivial equilibrium, the extinct equilibrium $(1, 0)$ is globally asymptotically stable (refer to Sect. 14.8). Even when a non-trivial equilibrium exists, the extinct equilibrium $(1, 0)$ is locally asymptotically stable if the parameter c/es is not so large such that $c/es < e_L/r$. When the parameter c/es is sufficiently large and a unique non-trivial equilibrium (E^*, L^*) exists with $E^* = E_s$, equilibrium (E^*, L^*) is globally asymptotically stable as seen in Fig. 5.14b.

Exercise 5.5 With the local stability analysis, show that the non-trivial equilibrium (E^*, L^*) is a locally asymptotically stable node (sink) or saddle when $m > 0$.

As a consequence, the population goes extinct when the immigration from the patches of large subpopulation to the empty patch is little active with small c , while it persists when it is sufficiently active with large c . However, even when the immigration from the patches of large subpopulation to the empty patch is little active, the population may persist with an initial condition if the immigration from the patches of large subpopulation to the patch of small subpopulation is sufficiently active. This is a bistable case as shown in Figs. 5.11 and 5.12b–d. The *rescue effect* called by Hanski [7] belongs to this case. In other words, the immigration from the patches of large subpopulation can work as a rescue effect to reduce the frequency of patches with small subpopulation, and subsequently it can decrease the probability of extinction.

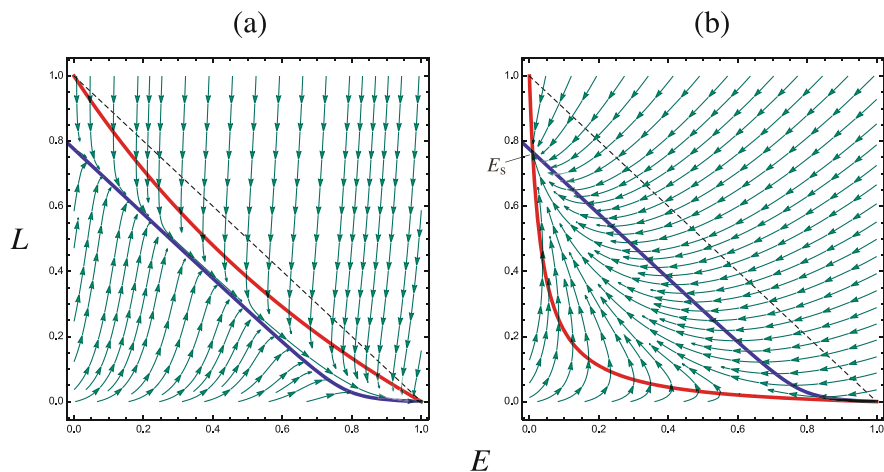


Fig. 5.14 Nullclines and vector flows in the (E, L) -phase plane. Numerically drawn for (a) $c = 0.05$ when there is no non-trivial equilibrium; (b) $c = 2.5$ when there is a unique non-trivial equilibrium. Commonly, $e_S = 0.08$; $e_L = 0.5$; $r = 0.02$; $m = 2.2$

Answer to Exercise

Exercise 5.1(p. 142)

The logistic equation (5.10) can be rewritten as

$$\frac{dN}{dt} = \left(\frac{r_0}{2\beta} \right)^2 - \beta \left(N - \frac{r_0}{2\beta} \right)^2.$$

The right side takes maximum when $N = r_0/2\beta$, that is, the net growth rate of population size dN/dt takes maximum $(r_0/2\beta)^2$ when the population size becomes the half saturation value $r_0/2\beta$.

Exercise 5.2 (p. 148)

As mentioned in Sect. 5.5, the recurrence relation (5.24) is mathematically equivalent to the logistic map (2.13) in Sect. 2.1.3 and (12.4) in Sect. 12.1.3. Thus, as described in Sects. 2.1.3 and 12.1.3, it is necessary for a monotonic approach to the equilibrium r_0/β that $1 < \mathcal{R}_0 = 1 + r_0\Delta t \leq 2$. Since $r_0 > 0$ and $\Delta t > 0$, this necessary condition leads to the following condition about the time step size Δt :

$$\Delta t \leq \frac{1}{r_0}. \quad (5.34)$$

We must remark that the recurrence relation (5.24) becomes nonsense as an approximation for the logistic equation for $\tilde{N}(0) \geq r_0/\beta + 1/(\beta\Delta t)$, since the subsequent sequence $\{\tilde{N}(\Delta t), \tilde{N}(2\Delta t), \tilde{N}(3\Delta t), \tilde{N}(4\Delta t), \dots\}$ becomes negative or all zero. Hence, we are going to consider the recurrence relation (5.24) only for the initial value $\tilde{N}(0) < r_0/\beta + 1/(\beta\Delta t)$.

Next, when $1 < \mathcal{R}_0 \leq 2$, the cobwebbing method (refer to Sect. 12.1.2) for the logistic map (12.4) shows that the orbit $\{\tilde{N}(0), \tilde{N}(\Delta t), \tilde{N}(2\Delta t), \tilde{N}(3\Delta t), \dots\}$ from the initial value $\tilde{N}(0) < r_0/\beta$ necessarily approaches equilibrium r_0/β in a monotonically increasing manner, as seen from Fig. 12.2b in Sect. 12.1.3, because the derivative of the graph at the intersection with the line of $x_{k+1} = x_k$, that is, at equilibrium r_0/β , is non-negative and less than 1. Therefore, the condition (5.34) is sufficient for the sequence $\{\tilde{N}(k\Delta t)\}$ to approach equilibrium r_0/β in a monotonically increasing manner when $\tilde{N}(0) < r_0/\beta$.

For the case where $r_0/\beta < \tilde{N}(0) < r_0/\beta + 1/(\beta\Delta t)$, the same arguments as the cobwebbing method on the logistic map (12.4) can be applied (see Fig. 12.2b). First, let us consider the case of $\mathcal{R}_0 = 1 + r_0\Delta t = 2$, that is, when $\Delta t = 1/r_0$. In this case, the graph of the logistic map has the intersection with the line of $x_{k+1} = x_k$ at the extremal maximum of the graph, so that the derivative is zero at the intersection. It can be easily found from the recurrence relation (5.24) that $0 < \tilde{N}(\Delta t) < r_0/\beta$ if $r_0/\beta < \tilde{N}(0) < r_0/\beta + 1/(\beta\Delta t)$. Thus, the subsequent sequence $\{\tilde{N}(\Delta t), \tilde{N}(2\Delta t), \tilde{N}(3\Delta t), \tilde{N}(4\Delta t), \dots\}$ must approach equilibrium r_0/β in a monotonically increasing manner, as already described in the above. As a result, when $\Delta t = 1/r_0$, we have $\tilde{N}(0) > \tilde{N}(\Delta t)$ and $\tilde{N}(k\Delta t) < \tilde{N}((k+1)\Delta t)$ for any $k > 0$. This means that the orbit from $\tilde{N}(0) \in (r_0/\beta, r_0/\beta + 1/(\beta\Delta t))$ does not monotonically approach r_0/β but does in a manner like the upper second figure in Fig. 2.9 (p. 43) of Sect. 2.1.3.

Next let us consider the case of $\mathcal{R}_0 = 1 + r_0\Delta t < 2$, that is, when $\Delta t < 1/r_0$. As exemplified in Fig. 12.2b, the graph of the logistic map has the intersection with the line of $x_{k+1} = x_k$ at a point in the left side of the extremal maximum of the graph. Thus, the cobwebbing method can show that $\tilde{N}(k\Delta t) < \tilde{N}((k-1)\Delta t)$ and $\tilde{N}(k\Delta t) > r_0/\beta$ for any $k > 0$ if

$$\frac{r_0}{\beta} < \tilde{N}(0) < \tilde{N}_c := \frac{1}{\beta\Delta t}, \quad (5.35)$$

where $\tilde{N}_c$ is derived from the root of the following equation of N such that $\tilde{N}_c > r_0/\beta$:

$$N + (r_0 - \beta N)N\Delta t = \frac{r_0}{\beta}.$$

It is the larger intersection of the graph of the logistic map with the horizontal line of the value r_0/β . As a result, when $\Delta t < 1/r_0$, if the condition (5.35) is satisfied, the sequence $\{\tilde{N}(k\Delta t)\}$ approaches r_0/β in a monotonically decreasing manner.

Further, it can be easily shown by the cobwebbing method that, when $\tilde{N}(0) > \tilde{N}_c$, we have $\tilde{N}(\Delta t) < r_0/\beta$, so that the orbit $\{\tilde{N}(k\Delta t)\}$ does not become monotonic but approaches r_0/β in a manner like the upper second figure in Fig. 2.9 (p. 43) of Sect. 2.1.3, as in the previous case mentioned above.

Lastly, let us consider the case of $\tilde{N}(0) = \tilde{N}_c$. In this case, we have $\tilde{N}(\Delta t) = r_0/\beta$ and $\tilde{N}(k\Delta t) = r_0/\beta$ for any $k > 0$. So the sequence $\{\tilde{N}(k\Delta t)\}$ can be regarded as monotonically decreasing in a mathematically wide sense.

Consequently, for the initial value $\tilde{N}(0) \in (r_0/\beta, r_0/\beta + 1/(\beta\Delta t))$, the condition (5.35) is necessary and sufficient for a monotonic approach of the sequence $\{\tilde{N}(k\Delta t)\}$ to the equilibrium r_0/β in a decreasing manner. Hence, from the condition (5.35), we have

$$\Delta t \leq \frac{1}{\beta\tilde{N}(0)}. \quad (5.36)$$

We note that $1/\{\beta\tilde{N}(0)\} < 1/r_0$ in this case, since this is for $\tilde{N}(0) > r_0/\beta$. Besides, when the condition (5.36) is satisfied, the initial value $\tilde{N}(0)$ is necessarily less than $r_0/\beta + 1/(\beta\Delta t)$.

As a result from the conditions (5.34) and (5.36), if and only if

$$\Delta t \leq \min \left[\frac{1}{\beta\tilde{N}(0)}, \frac{1}{r_0} \right] = \frac{1}{r_0} \min \left[\frac{1}{\beta\tilde{N}(0)/r_0}, 1 \right], \quad (5.37)$$

the sequence $\{\tilde{N}(k\Delta t)\}$ generated by the recurrence relation (5.24) becomes monotonic to approach r_0/β .

Exercise 5.3 (p. 149)

As mentioned in Sect. 5.5, the recurrence relation (5.25) is mathematically equivalent to Ricker model (2.7) in Sect. 2.1.2. We shall now apply the logically same arguments as those in Exercise 5.2.

As described in Sect. 2.1.2, it is necessary for a monotonic approach to the equilibrium r_0/β that $1 < \mathcal{R}_0 = e^{r_0\Delta t} \leq e$. Since $r_0 > 0$ and $\Delta t > 0$, this necessary condition leads to the condition about the time step size Δt (5.34) same as for Exercise 5.2 about the recurrence relation (5.24).

Next, when $1 < \mathcal{R}_0 \leq e$, by the same arguments as in Exercise 5.2 with the cobwebbing method (refer to Sect. 12.1.2) for the logistic map (12.4), it can be shown that the orbit $\{\tilde{N}(0), \tilde{N}(\Delta t), \tilde{N}(2\Delta t), \tilde{N}(3\Delta t), \dots\}$ from the initial value $\tilde{N}(0) < r_0/\beta$ necessarily approaches equilibrium r_0/β in a monotonically increasing manner (refer to Fig. 5.15). Therefore, the condition (5.34) is sufficient for an approach of the sequence $\{\tilde{N}(k\Delta t)\}$ from $\tilde{N}(0) < r_0/\beta$ to equilibrium r_0/β in a monotonically increasing manner.

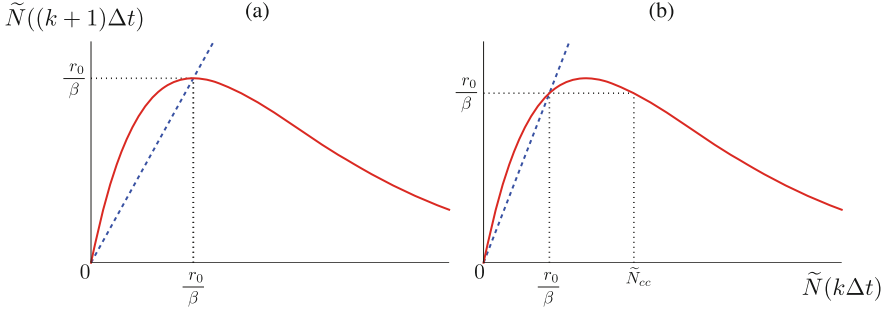


Fig. 5.15 The illustrative graph of Ricker model (5.25). (a) $\Delta t = 1/r_0$; (b) $\Delta t = 1/r_0$.

For $\tilde{N}(0) > r_0/\beta$, let us first consider the case of $\mathcal{R}_0 = e^{r_0\Delta t} = e$, that is, when $\Delta t = 1/r_0$, as in Exercise 5.2. In this case, as seen in Fig. 5.15a, the graph of Ricker model has the intersection with the line of $N_{k+1} = N_k$ at the extremal maximum of the graph, so that the derivative is zero at the intersection. It can be easily found from the recurrence relation (5.25) that $\tilde{N}(\Delta t) < r_0/\beta$ if $\tilde{N}(0) > r_0/\beta$. Thus, the subsequent sequence $\{\tilde{N}(\Delta t), \tilde{N}(2\Delta t), \tilde{N}(3\Delta t), \tilde{N}(4\Delta t), \dots\}$ must approach equilibrium r_0/β in a monotonically increasing manner, as already described in the above. As a result, when $\Delta t = 1/r_0$, we have $\tilde{N}(0) > \tilde{N}(\Delta t)$ and $\tilde{N}(k\Delta t) < \tilde{N}((k+1)\Delta t)$ for any $k > 0$. This means that the orbit does not monotonically approach r_0/β from $\tilde{N}(0) > r_0/\beta$.

Next let us consider the case of $\mathcal{R}_0 = e^{r_0\Delta t} < e$, that is, when $\Delta t < 1/r_0$. In this case, the graph of Ricker model has the intersection with the line of $N_{k+1} = N_k$ at a point in the left side of the extremal maximum of the graph, as seen in Fig. 5.15b. Thus, the cobwebbing method can show that $\tilde{N}(k\Delta t) < \tilde{N}((k-1)\Delta t)$ and $\tilde{N}(k\Delta t) > r_0/\beta$ for any $k > 0$ if

$$\frac{r_0}{\beta} < \tilde{N}(0) < \tilde{N}_{cc}, \quad (5.38)$$

where $\tilde{N}_{cc}$ is the root of the following equation of N such that $\tilde{N}_{cc} > r_0/\beta$:

$$Ne^{(r_0 - \beta N)\Delta t} = \frac{r_0}{\beta}.$$

It is the larger intersection of the graph of Ricker model with the horizontal line of the value r_0/β . As a result, when $\Delta t < 1/r_0$, if the condition (5.38) is satisfied, the sequence $\{\tilde{N}(k\Delta t)\}$ approaches r_0/β in a monotonically decreasing manner.

Further, it can be easily shown by the cobwebbing method that, when $\tilde{N}(0) > \tilde{N}_{cc}$, we have $\tilde{N}(\Delta t) < r_0/\beta$, so that the orbit $\{\tilde{N}(k\Delta t)\}$ does not become monotonic.

Lastly, let us consider the case of $\tilde{N}(0) = \tilde{N}_{cc}$. In this case, we have $\tilde{N}(\Delta t) = r_0/\beta$ and $\tilde{N}(k\Delta t) = r_0/\beta$ for any $k > 0$. So the sequence $\{\tilde{N}(k\Delta t)\}$ can be regarded as monotonically decreasing in a mathematically wide sense.

Consequently, for the initial value $\tilde{N}(0) > r_0/\beta$, the condition (5.38) is necessary and sufficient for a monotonic approach of the sequence $\{\tilde{N}(k\Delta t)\}$ to equilibrium r_0/β in a decreasing manner. The condition (5.38) is mathematically equivalent to

$$\tilde{N}(0)e^{\{r_0 - \beta\tilde{N}(0)\}\Delta t} \geq \frac{r_0}{\beta}$$

as easily seen from Fig. 5.15b. That is, we find the following condition in terms of Δt :

$$\Delta t \leq \frac{1}{r_0} \frac{\ln[\beta\tilde{N}(0)/r_0]}{\beta\tilde{N}(0)/r_0 - 1}. \quad (5.39)$$

It can be easily shown that the right side of (5.39) is greater than $1/r_0$ for $\tilde{N}(0) < r_0/\beta$, and less than $1/r_0$ for $\tilde{N}(0) > r_0/\beta$, while it converges to $1/r_0$ as $\tilde{N}(0) \rightarrow r_0/\beta$.

As a result from the conditions (5.34) and (5.39), if and only if

$$\Delta t \leq \frac{1}{r_0} \min \left[\frac{\ln[\beta\tilde{N}(0)/r_0]}{\beta\tilde{N}(0)/r_0 - 1}, 1 \right], \quad (5.40)$$

the sequence $\{\tilde{N}(k\Delta t)\}$ generated by the recurrence relation (5.25) becomes monotonic to approach r_0/β .

Now we shall compare the recurrence relation (5.25) with (5.24) with respect to which could be regarded as the better approximation for the logistic equation (5.10). Since the recurrence relation (5.24) can become an approximation for the logistic equation only for the initial value $\tilde{N}(0) \in (0, r_0/\beta + 1/(\beta\Delta t))$, let us compare the recurrence relation (5.25) with (5.24) according to the condition for the monotonicity about the sequence generated by the recurrence relation only for $\tilde{N}(0) \in (0, r_0/\beta + 1/(\beta\Delta t))$.

For $\tilde{N}(0) \in (0, r_0/\beta]$, the obtained conditions (5.37) and (5.40) for the recurrence relations (5.24) and (5.25) respectively are identically given by (5.34). Hence for $\tilde{N}(0) \in (0, r_0/\beta]$, they could be regarded as compatible with respect to the monotonicity of the sequence generated by them.

For $\tilde{N}(0) > r_0/\beta$, we can prove that the following equality holds:

$$\frac{1}{\beta\tilde{N}(0)/r_0} < \frac{\ln[\beta\tilde{N}(0)/r_0]}{\beta\tilde{N}(0)/r_0 - 1}.$$

Therefore, for $\tilde{N}(0) \in (r_0/\beta, r_0/\beta + 1/(\beta\Delta t))$, the condition (5.40) is weaker than (5.37). Moreover, the condition (5.40) is applicable also for $\tilde{N}(0) \geq r_0/\beta + 1/(\beta\Delta t)$. That is, the recurrence relation (5.25) can be used as an approximation for the logistic equation even for $\tilde{N}(0) \geq r_0/\beta + 1/(\beta\Delta t)$ under the condition (5.40).

As a consequence from these arguments, it is implied that the recurrence relation (5.25) could be regarded as better than (5.24) according to the monotonicity of the sequence generated by the recurrence relation.

Exercise 5.4 (p. 155)

For the non-trivial equilibrium (E^*, L^*) of the system (5.31), we can derive the linearized system around (E^*, L^*) with the following Jacobian matrix A (refer to Sect. 14.2):

$$A = \begin{pmatrix} -a^* & -b^* \\ -c^* & -d^* \end{pmatrix}, \quad (5.41)$$

where a^* , b^* , c^* , and d^* are positive elements defined as

$$\begin{aligned} a^* &:= e_S + cL^*; & b^* &:= e_S + cE^*; & c^* &:= r + mL^*; \\ d^* &:= 2mL^* + r + e_L - m(1 - E^*) = r + mL^* + \frac{re_L}{r + mL^*}. \end{aligned}$$

As described in Sect. 14.3, we can determine the local stability of (E^*, L^*) by the eigenvalues of A . The characteristic equation to determine the eigenvalue λ for A becomes

$$\lambda^2 + (a^* + d^*)\lambda + a^*d^* - b^*c^* = 0. \quad (5.42)$$

For the equilibrium (5.33) when $m = 0$, the characteristic equation (5.42) becomes

$$\lambda^2 + (e_S + cL^* + r + e_L)\lambda + cr(1 - E^*) = 0.$$

When the equilibrium (5.33) exists, we can easily find that the discriminant for this characteristic equation is necessarily positive, and two eigenvalues are different negative values, making use of the relation between the roots and coefficients. Therefore, the equilibrium (5.33) is locally asymptotically stable as a stable node, that is, a sink when it exists.

Exercise 5.5 (p. 158)

About Jacobian matrix A for the non-trivial equilibrium (E^*, L^*) of (5.31), given by (5.41) shown in the answer for Exercise 5.4, we have $\text{tr } A = a^* + d^* > 0$. Thus, by the relation between the roots and coefficients of the characteristic equation (5.42), if the eigenvalue is imaginary only when $\det A = a^*d^* - b^*c^* > 0$, then the real part must be negative. If the eigenvalue is real, both eigenvalues are negative when $\det A > 0$, while they have different signs when $\det A < 0$. Therefore, when a non-trivial equilibrium (E^*, L^*) exists for (5.31), it is a locally asymptotically stable node (sink) if $\det A > 0$, while it is unstable as a saddle if $\det A < 0$. This result implies that, in a bistable case with two non-trivial equilibria, one is a sink and the other is a saddle.

References

1. N. Bacaër, *A Short History of Mathematical Population Dynamics* (Springer, London, 2011)
2. H. Fujita, S. Utida, The effect of population density on the growth of an animal population. *Ecology* **34**(3), 488–498 (1953)
3. M.E. Gilpin, F.J. Ayala, Global models of growth and competition. *Proc. Natl. Acad. Sci. U.S.A.* **70**, 3590–3593 (1973)
4. B. Gompertz, On the nature of the function expressive of the law of human mortality and on a new mode of determining life contingencies. *Philos. Trans. Roy. Soc. Lond. A* **115**, 513–585 (1825)
5. I. Hanski, Single-species spatial dynamics may contribute to long-term rarity and commonness. *Ecology* **66**, 335–343 (1985)
6. I. Hanski, Single-species metapopulation dynamics: concepts, models and observations. *Biol. J. Linnean Soc.* **42**, 17–38 (1991)
7. I. Hanski, *Metapopulation Ecology*. Oxford Series in Ecology and Evolution (Oxford University Press, Oxford, 1999)
8. F. John, *Lectures on Advanced Numerical Analysis* (Gordon and Breach, Science Publishers, New York, 1967)
9. E. Kreyszig, *Advanced Engineering Mathematics*, 10th edn. (Wiley, New York, 2011)
10. R. Levins, Some demographic and genetic consequences of environmental heterogeneity for biological control. *Bull. Entomol. Soc. Am.* **15**, 237–240 (1969)
11. R. Levins, Extinction, in: *Some Mathematical Problems in Biology*, ed. by M. Gerstenhaber, *Lectures on Mathematics in the Life Sciences*, vol. 2 (American Mathematical Society, Rhode Island, 1970), pp. 75–107
12. H. Seno, Some time-discrete models derived from ODE for single-species population dynamics: Leslie's idea revisited. *Scientiae Mathematicae Japonicae* **58**(2), 389–398 (2003)
13. F.E. Smith, Population dynamics in *Daphnia magna* and a new model for population growth. *Ecology* **44**, 651–663 (1963)
14. P.F. Verhulst, Notice sur la loi que la population suit dans son accroissement. *Correspond. Math. Phys.* **10**, 113–121 (1838)
15. P.F. Verhulst, Recherches mathématiques sur la loi d'accroissement de la population. *Nouv. mém. de l'Académie Royale des Sci. et Belles-Lettres de Bruxelles* **18**, 1–41 (1845)
16. S. Wright, Review of *The Biology of Population Growth*, by Raymond Pearl. *J. Amer. Stat. Assoc.* **21**, 493–497 (1926)